

Lab 01 manual

Introduction to Web Engineering

Web Engineering is the application of systematic and quantifiable approaches (concepts methods, techniques tools) to cost - effective requirements analysis, design, implementation, testing, operation, and maintenance of high quality Web applications.

Technologies to be studied

- HTML5
- CSS3
- JAVASCTIPY
- Bootstrap5
- React.js
- Asp.Net
- Add.net
- MVC
- EF
- Api

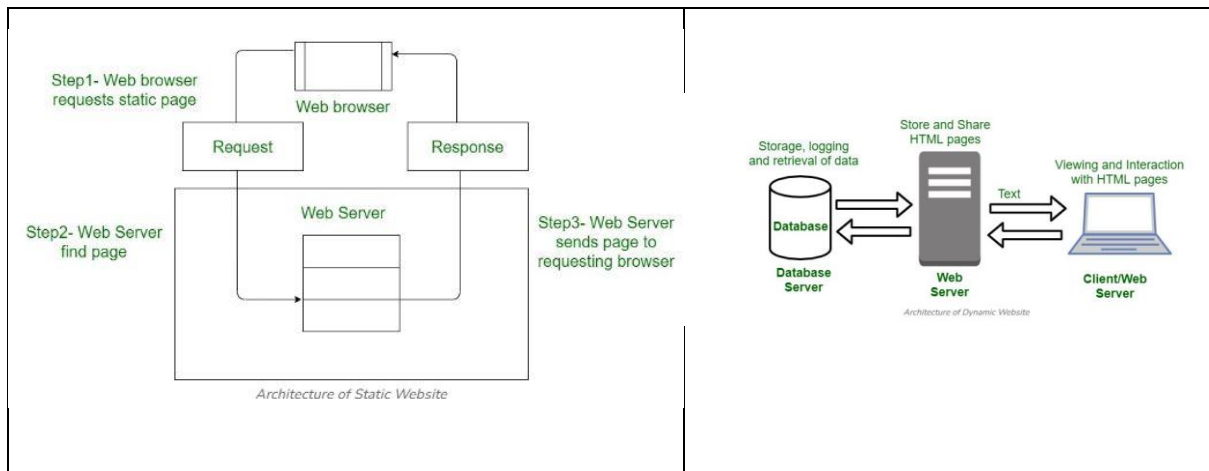
Tools – IDEs

Visual studio

TYPES OF WEBSITES

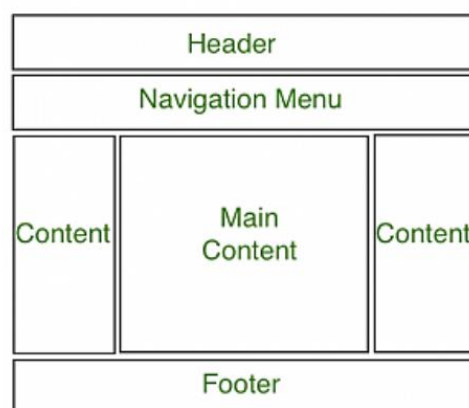
When we type a certain URL in a browser search bar, the browser requests the page from the Web server and the Web server returns the required web page and its content to the browser. Now, it differs how the server returns the information required in the case of static and dynamic websites.

Static websites	Dynamic websites
Content of Web pages cannot be change at runtime.	Content of Web pages can be changed.
No interaction with database possible.	Interaction with database is possible
Cheaper Development costs.	More Development costs
HTML, CSS, JavaScript is used for developing the website.	Server-side languages such as PHP, Node.js are used.



Web structure

Website structure refers to the organizational design of a website's pages. Creating a website structure involves organizing and classifying content to create a layout that's understandable, accessible, and predictable. Navigation should be intuitive so users can find what they're looking for on any page without much effort. A clear and well-structured website allows users and search engines to easily find and understand the information they need. It is important to structure your individual website pages or blog posts in a way that makes it easy for users to find what they are looking for, find similar content and understand where they are on your website. Breadcrumb trails, tags, and contextual site links are used to structure information architecture on individual pages. Take care of the headers that you put on individual pages. Make sure that they follow the right order, for example, the title of the blog post is H1 and that they all have metadata. Metadata are important part of UX, too. You don't want to confuse users what your site is about



install VS Code including extension

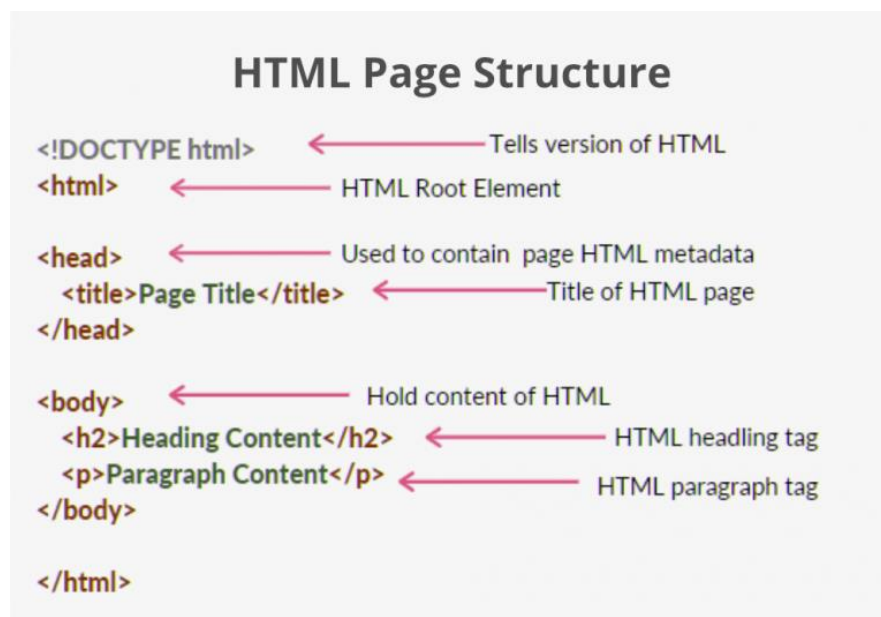
- Visit the site <https://code.visualstudio.com/download>
- Then choose based on your computer windows types supporting VS-Code.
- Add extension according to your requirement

WHAT IS HTML?

HTML stands for Hypertext Markup Language. It is the standard language used to create and structure content on the web. It tells the web browser how to display text, links, images, and other forms of multimedia on a webpage. HTML sets up the basic structure of a website.

HTML Page Structure

HTML has essential building-block elements (i.e. doctype declaration, HTML, head, title, and body elements) upon which all web pages are created.



- The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the HTML page

- The <title> element specifies a title for the HTML page (which is shown in
- the browser's title bar or in the page's tab)
- The <body> element defines the document's body, and is a container for all
- the visible contents, such as headings, paragraphs, images, hyperlinks,
- tables, lists, etc.
- The <h1> element defines a large heading
- The <p> element defines a paragraph

HTML History:

Since the early days of the World Wide Web, there have been many versions of HTML:

Year	Version
1989	Tim Berners-Lee invented www
1991	Tim Berners-Lee invented HTML
1993	Dave Raggett drafted HTML+
1995	HTML Working Group defined HTML 2.0
1997	W3C Recommendation: HTML 3.2
1999	W3C Recommendation: HTML 4.01
2000	W3C Recommendation: XHTML 1.0
2008	WHATWG HTML5 First Public Draft
2012	WHATWG HTML5 Living Standard
2014	W3C Recommendation: HTML5
2016	W3C Candidate Recommendation: HTML 5.1
2017	W3C Recommendation: HTML5.1 2nd Edition
2017	W3C Recommendation: HTML5.2

Web Browsers

The purpose of a web browser (Chrome, Edge, Firefox, Safari) is to read HTML documents and display them correctly. A browser does not display the HTML tags, but uses them to determine how to display the document.

HTML Tags

HTML has two types of tags: container tags (e.g., <h1></h1>, <p></p>) and standalone tags (e.g., , <input>)

HTML Heading

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>This is a Heading</h1>
<h2>This is a Heading</h2>

<h3>This is a Heading</h3>
<h4>This is a Heading</h4>

<h5>This is a Heading</h5>

<h6>This is a Heading</h6>
</body>
</html>
```

HTML Paragraph

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

The HTML `<br>` element defines a line break.

The `<hr>` tag is an empty tag, which means that it has no end tag.

The HTML `<pre>` element defines preformatted text.

The text inside a `<pre>` element is displayed in a fixed-width font (usually Courier), and it preserves both spaces and line breaks

HTML Link using Anchor Tag

```
<a href="https://www.w3schools.com">This is a link</a>
```

HTML Image

```

```

Link Addressing

- Absolute address
- Relative address

HTML List

HTML List Tags

Tag	Description
<code></code>	Defines an unordered list
<code></code>	Defines an ordered list
<code></code>	Defines a list item
<code><dl></code>	Defines a description list
<code><dt></code>	Defines a term in a description list
<code><dd></code>	Describes the term in a description list

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

<dl>
  <dt>Coffee</dt>
  <dd>Black hot drink</dd>
  <dt>Milk</dt>
  <dd>White cold drink</dd>
</dl>
```

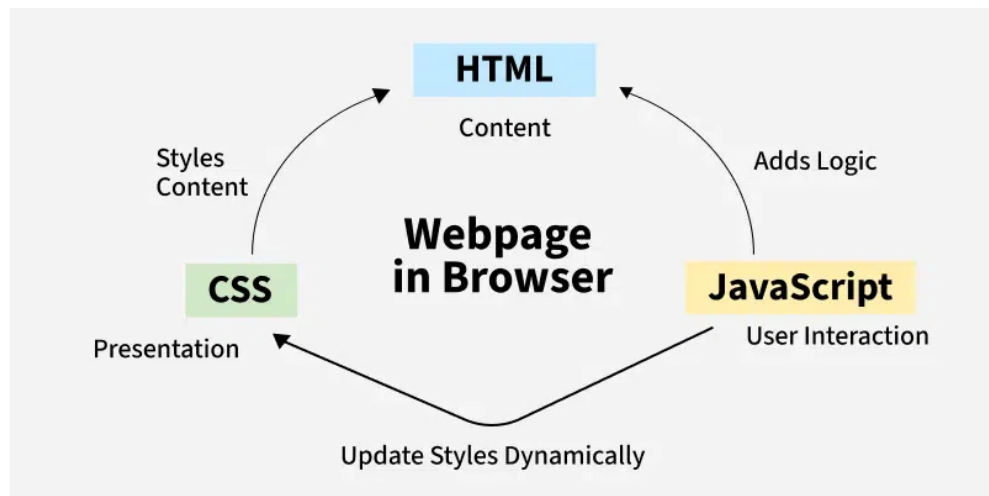
INTRODUCTION OF CSS

CSS stands for Cascading Style Sheets. It is a stylesheet language used to style and enhance website presentation.

CSS is one of the main three components of a webpage along with HTML and JavaScript.

HTML adds Structure to a Webpage, JavaScript adds logic to it and CSS makes it visually appealing or stylish. It controls the layout of a web page i.e. how HTML elements will be displayed on a webpage.

CSS was released (in 1996), 3 years after HTML (in 1993). The main idea behind its use is, it allows the separation of content (HTML) from presentation (CSS). This makes websites easier to maintain and more flexible.



There are three different ways to add CSS styles to an HTML document.

Inline CSS

Use the style attribute on the HTML element to add styles to the web page. It is used for small projects.

```
<!-- File name: index.html -->
<html>
<body>
  <!-- Using Inline CSS -->
  <h3 style="text-align: center;">
    Welcome To GFG
  </h3>

  <p>CSS Tutorial - GeeksforGeeks</p>
</body>
</html>
```

Internal CSS

Place the CSS styles within a <style> tag inside the HTML file, usually inside the <head> section.

```
<!-- File name: index.html -->
<html>
<head>
  <!-- Using Internal CSS -->
  <style>
    h3 {
      color: green;
    }
  </style>
</head>
<body>
  <!-- CSS is applied here -->
  <h3>Welcome To GFG</h3>

  <p>CSS Tutorial - GeeksforGeeks</p>
</body>
</html>
```

External CSS

Create a separate CSS file with a .css extension and link this file to your HTML file using the <link> tag.

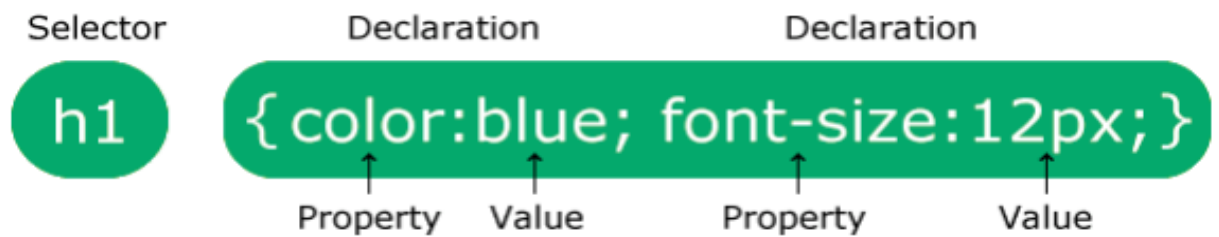
```
<!-- File name: index.html -->
<html>
<head>
  <!-- Importing External CSS -->
  <link rel="stylesheet" href="style.css" />
</head>
<body>
  <p>CSS Tutorial - GeeksforGeeks</p>
</body>
</html>
```

```
/* Write CSS Here */ /* External CSS */
/* File name: style.css */
p {
  text-align: center;
}
```


CSS SYNTAX:

All CSS is done in style tag for internal:

`<style> </style>`



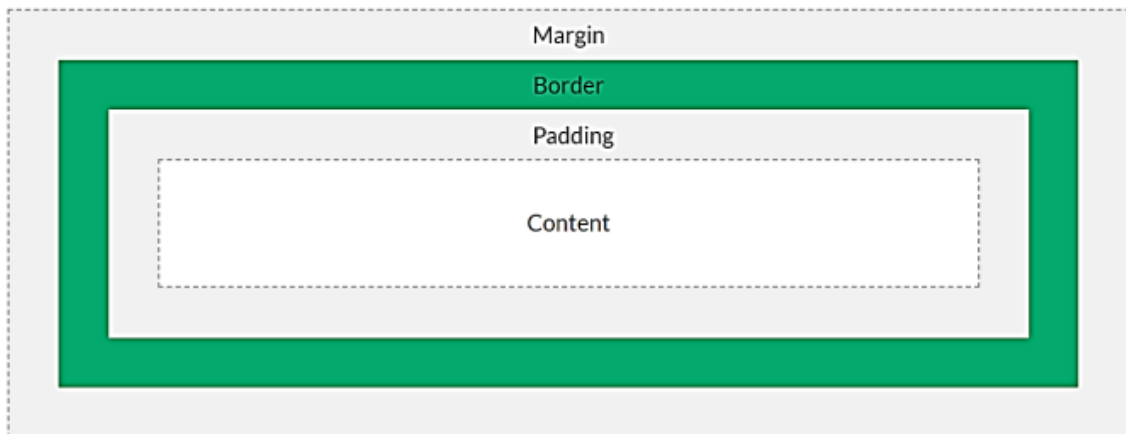
CSS SELECTORS:

CSS selectors are used to select the content you want to style. Selectors are the part of CSS rule set. CSS selectors select HTML elements according to its id, class, type, attribute etc.

There are several different types of selectors in CSS.

1. CSS Element Selector
2. CSS Id Selector #
3. CSS Class Selector .
4. CSS Universal Selector *
5. CSS Group Selector used with , commas

CSS Box Model: [Padding Vs Margin];

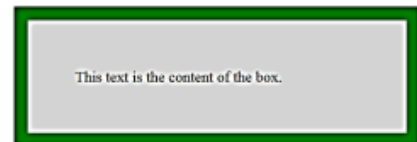


- **Content** - The content of the box, where text and images appear
- **Padding** - Clears an area around the content. The padding is transparent
- **Border** - A border that goes around the padding and content
- **Margin** - Clears an area outside the border. The margin is transparent

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  background-color: lightgrey;
  width: 300px;
  border: 15px solid green;
  padding: 50px;
  margin: 20px;
}
</style>
</head>
<body>

<div>This text is the content of the box.</div>

</body>
</html>
```



Positions

The position CSS property sets how an element is positioned in a document. The top, right, bottom, and left properties determine the final location of positioned elements.

There are five different position values:

- static.
- relative.
- fixed.

- absolute.
- sticky

```
position: static;
```

```
position: relative;  
top: 40px; left: 40px;
```

```
position: absolute;  
top: 40px; left: 40px;
```

```
position: sticky;  
top: 20px;
```

In this demo you can control the `position` property for the yellow box.



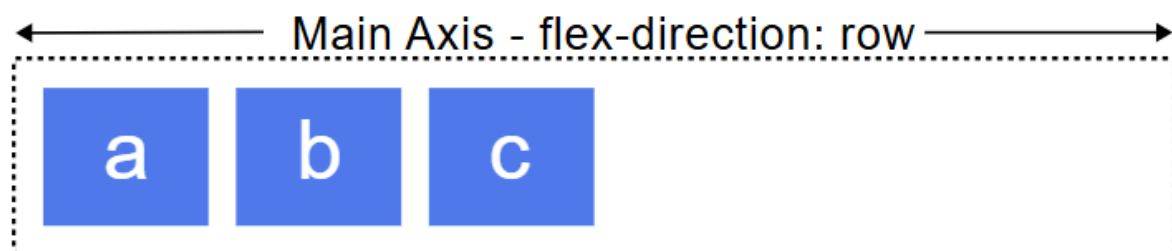
To see the effect of sticky positioning, select the `position: sticky` option and scroll this container.

The element will scroll along with its container, until it is at the top of the

Flex container

The flexible box layout module (usually referred to as flexbox) is a one-dimensional layout model for distributing space between items and includes numerous alignment capabilities.

When we describe flexbox as being one-dimensional we are describing the fact that flexbox deals with layout in one dimension at a time — either as a row or as a column.



Main properties of flex

Display: flex;

Justify-content: center;

Align-item: center;

Flex-wrap: wrap;

Task1

Create a document using HTML tags based on the provided sample.

Pakistan Zindabad
Independence Day (Urdu: مہِ آزادی <i>Yaum-e-Azadi</i>), observed annually on 14 August, is a national holiday in Pakistan it commemorates the day when Pakistan achieved independence and was declared a sovereign state following the end of the British Raj in 1947. On 14 August 1947, the new Dominion of Pakistan became independent and Muhammad Ali Jinnah was sworn in as its first governor general in Karachi. Independence was marked with widespread celebration, but the atmosphere remained heated given the communal riots prevalent during <i>independence in 1947</i>
Provinces of Pakistan
there are 5 Provinces of Pakistan
- Sindh - Punjab - Balochistan - Khyber Pakhtunkhwa - Gilgit Baltistan
Sindh
Sindh (/ˈsɪnd/; Sindhi: سنڌ; Urdu: سینڈھ; historically romanized as <i>Sind</i>) is one of the four provinces of Pakistan. Located in the southeastern region of the country, Sindh is the third-largest province of Pakistan by total area and the second-largest province by population after Punjab. It shares land borders with the Pakistani provinces of Balochistan and Punjab to the north, respectively, and the Indian states of Gujarat and Rajasthan to the east; it is also bounded by the Arabian Sea to the south. Sindh's landscape consists mostly of alluvial plains flanking the Indus River, the Thar Desert in the eastern portion of the province along the international border with India, and the Kirthar Mountains in the western portion of the province.
Punjab
Punjab (Punjabi and Urdu: پنجاب, <i>Panjab</i>) is one of the four provinces of Pakistan. Punjab is the second-largest province of Pakistan by land area and the most populous Pakistani province. Forming the bulk of the transnational Punjab region between Pakistan and India, it is bounded locally by Sindh to the south, Balochistan to the west, Khyber Pakhtunkhwa to the northwest, the Islamabad Capital Territory to the north, and the Pakistan-administered territory of Azad Jammu and Kashmir to the northeast. On its eastern side, it is bounded by the India–Pakistan border, sharing an international boundary with the Indian states of Punjab and Rajasthan to the east and southeast, respectively, and a disputed boundary with the Indian-administered territory of Jammu and Kashmir to the northeast. The province's capital is Lahore—a cultural, historical, economic, and cosmopolitan centre of Pakistan.
Balochistan
Balochistan (Balochi: بلوچستان) is one of the four provinces of Pakistan. It is the largest province in terms of land area, forming the southwestern region of the country, but is the least populated. Its provincial capital and largest city is Quetta. Balochistan shares borders with Punjab and the Khyber Pakhtunkhwa to the northeast, Sindh to the east and southeast, the Arabian Sea to the south, Iran (Sistan and Baluchestan) to the west and Afghanistan (Helmand, Nimruz, Kandahar, Paktika and Zabul Provinces) to the north and northwest.
Khyber Pakhtunkhwa
Khyber Pakhtunkhwa (Urdu: خیبر پختونخوا), commonly abbreviated as KP or KPK, is one of the four provinces of Pakistan. It is located in the northwestern region of the country, along the Afghanistan–Pakistan border, and the city of Peshawar serves as its capital. While it is the third-largest Pakistan province in terms of both its population and its economy, it is geographically the smallest.
Gilgit Baltistan
Gilgit-Baltistan (Urdu: گلگت-بلتستان), formerly known as the Northern Areas, is a region administered by Pakistan as an autonomous territory, and constituting the northern portion of the larger Kashmir region which has been the subject of a dispute between India and Pakistan since 1947, and between India and China from somewhat later. It borders Azad Kashmir to the south, the province of Khyber Pakhtunkhwa to the west, the Wakhan Corridor of Afghanistan to the north, the Xinjiang region of China, to the east and northeast, and the Indian-administered union territories Jammu and Kashmir and Ladakh to the southeast.
LONG LIVE PAKISTAN

Task2

The objective of this lab task is to learn how to implement a hyper link and how to insert an image on a web page.

- Create New HTML Page
- Add Paragraphs
- Make some content of paragraph as Bold, Italic and Underline
- Insert a hyper link.
- Insert an image.

Task3

Create a navigation menu with nested dropdowns.

- Include a mega menu section for one of the dropdowns.
- Use CSS to style the dropdown with smooth animations.

Task4

Using position Create a card or box with multiple overlapping layers, ensuring the third layer is displayed as the top priority.

Task5

Create a web page that display the below table;

```

<html>
<head>
<title>This Is Your Title </title>
</head>
<body>
<h1>This Is Your Header </h1>
<p>This is your paragraph. </p>
</body>
</html>

```

Task6

Create a layout using flex container according to given sample

Top

Prev

Next

Bottom

Search

Delete

Save

Edit

New

Bill Reference

Exit

To :

Party Name :

Date :

Gold Type:

Gold Recieve :

dd/mm/yyyy

Teezabi

SEND

gold

cash

CONVERT TO TEEZABI

CONVERT TO PIECE

CUSTOMER INFORMATION

Name

Task7

Create a webpage that display your basic CV, using the tags which you learn in this lab.

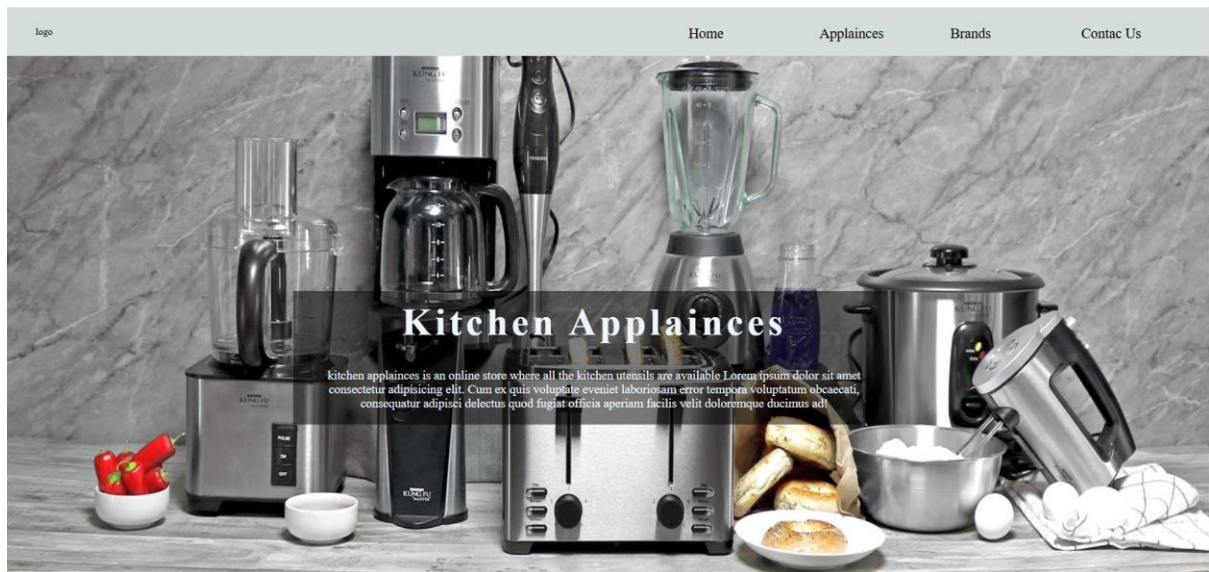
Task8

Create a grid layout using flex container according to given sample



Task9

Create a webpage for kitchen appliances that includes a navbar with links for logo, home, appliances, brands, and contact. Each link should lead to related content, such as images, text, and descriptions, and the page should also feature a footer.
given sample are:



Products



Brands



Payment Method



TASHA

We provide our best service to you. Lorem ipsum, dolor sit amet consectetur adipiscing elit. Voluptatibus quae incidunt quaerat mmm

FOLLOW US



Live Contact



Using an HTML table, create a cart layout for products based on the task mentioned above